

Extensions of the Quaternionic Flavor Framework: Sterile Neutrinos, QFT, and Gravity

The exceptional-geometric flavor framework, rooted in octonionic algebra and the Quaternionic Kernel Theorem (QKT), offers a compelling pathway for extending the Standard Model to incorporate sterile neutrinos, integrate with Quantum Field Theory (QFT), and potentially unify with gravity. The inherent algebraic structure, particularly the non-associativity of octonions and the symmetries of the exceptional Jordan algebra, provides a rich landscape for these theoretical advancements.

1. Sterile Neutrinos

The framework naturally accommodates sterile neutrinos, which are often identified with fundamental elements within the octonionic structure:

- **Fock Vacuum Identification:** In some octonionic models, the Standard Model gauge group is described as a subgroup that preserves the sterile neutrino, which is identified with the Fock vacuum [5](#). This suggests a fundamental, rather than ad-hoc, role for sterile neutrinos within the algebraic construction.
- **Left-Right Symmetric Frameworks:** The exceptional Jordan algebra $J_3(\mathbb{O}_\mathbb{C})$ is utilized in theories like the $E_8 \times \omega E_8$ unification, which naturally includes a left-right symmetric framework with sterile neutrinos [2](#). This implies that the algebraic structure itself can dictate the existence and properties of these additional neutral states.
- **Expanded Neutral Sector:** As previously discussed in the context of the E_6 inspired model, the neutral sector is expanded to include right-handed neutrinos (ν^c) and E_6 singlets (s), which are the sterile components. The QKT, by defining the fundamental algebraic blocks, provides the underlying geometric reason for their inclusion and their distinct mixing properties from active neutrinos.

2. Quantum Field Theory (QFT)

The non-associativity of octonions presents both a challenge and an opportunity for QFT, leading to the development of non-associative QFTs:

- **Non-Associative QFT:** The unique non-associative nature of octonions is directly being explored to formulate new QFTs [6](#) [7](#). This involves developing perturbative expansions and computing polarization correlation patterns induced by the octonion

structure constants. This suggests that the algebraic properties of octonions could lead to novel predictions for quantum phenomena.

- **Octonion Kernels in QFT:** Octonion kernels are being used to transform octonion-valued signals into the frequency domain within a QFT context, addressing limitations of conventional QFT [8](#). This indicates a potential for octonionic methods to enhance the mathematical tools of QFT.
- **Pre-Quantum, Pre-Spacetime Theories:** The framework extends to pre-quantum, pre-spacetime theories where the definition of spin leads to an eight-dimensional octonionic spacetime [9](#). This suggests a deeper connection between the algebraic structure and the fundamental nature of reality at the quantum level.

3. Gravity

The integration of octonionic structures with gravity is a more ambitious, but actively explored, area of research:

- **Octonionic Spacetime Geometry:** Some theories propose enlarging ordinary spacetime coordinates to octonionic-valued coordinates, leading to an octonionic geometry of gravity [10](#). This could provide a framework for understanding gravity at a more fundamental level, potentially addressing issues with quantum gravity.
- **Gravitational Instantons and Supergravity:** Octonionic gravitational instantons have been constructed by solving self-duality equations for the spin-connection [11](#). Furthermore, the roles of normed division algebras (including octonions) are made manifest in various supergravity theories, suggesting a deep algebraic foundation for these advanced gravitational theories [12](#).
- **Quantum Gravity:** The framework is being explored in the context of quantum gravity, with some approaches deriving a theory of quantum gravity containing an AdS_3 isometry/qubit duality that is isomorphic to the complexified octonion algebra [13](#). This indicates that octonions might provide the mathematical language necessary to unify quantum mechanics and general relativity.
- **Emergent Gravity:** Some theories propose that gravitation and quantum theory emerge from an $E_8 \times \omega E_8$ theory of unification, which utilizes octonionic triality and the exceptional Jordan algebra [2](#). This suggests that gravity itself could be an emergent phenomenon from the underlying octonionic structure.

4. Conclusion: A Unified Algebraic Foundation

The quaternionic flavor framework, built upon octonionic algebra and the Quaternionic Kernel Theorem, offers a powerful and consistent mathematical language for extending our

understanding of fundamental physics. It provides natural explanations for:

- **Sterile Neutrinos:** Their existence and properties are inherent to the expanded neutral sector dictated by the algebraic structure.
- **Quantum Field Theory:** The non-associativity of octonions opens new avenues for formulating QFTs that could lead to novel predictions.
- **Gravity:** The framework offers promising directions for unifying gravity with quantum mechanics, potentially leading to an emergent understanding of spacetime and fundamental interactions.

This approach suggests that the fundamental constants and symmetries of nature, including those governing flavor, QFT, and gravity, may ultimately be derived from the intrinsic properties of exceptional algebraic structures.

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